

Natural Language Processing

Welcome to the Natural Language Processing (NLP) learning roadmap. The objective of this module is to guide you from the foundational mechanics of text processing to advanced deep learning architectures for sequential data. **It is crucial that you complete these modules in the exact order presented.** NLP is a highly cumulative field—you must understand how to clean and prepare text before you can vectorize it (Bag of Words/TF-IDF), and a solid grasp of basic statistical models (N-grams) is required before you can successfully tackle complex neural networks like RNNs and LSTMs. Follow this sequence closely to build a robust, intuitive understanding of how machines interpret human language.

Key Topics you will Learn

- **Language and Knowledge:** How language encodes and transmits knowledge; ambiguity and context
- **Phase of NLP: Phonological / Phonetic: Lexical Syntactic Semantic Pragmatic: Discourse:**
- **Morphology and POS Tagging:** Tokenization, stemming, lemmatization; rule-based vs ML taggers.
- **Lexical Semantics:** Word sense disambiguation; WordNet; synonymy, hyponymy, polysemy
- **Text Preprocessing:** Cleaning Pipelines: lowercasing, stop-word removal, normalization
- **TF-IDF :** Term frequency-inverse document frequency; implementation with scikit-learn
- **Naive Bayes for Text:** Bag of words representation; Multinomial NB
- **Language Model:** N-gram Model, Hidden Markov Model, genisim, spacy
- **RNNs and Long-Range Dependencies:** Vanishing gradients in RNNs; LSTM/GRU overview; handling long sentences
- **RNN vs CNN for NLP:** Sequence Modelling vs local feature extraction; when to use which

Learning Roadmap

1. Phases of NLP:
 - a. [Phases of Natural Language Processing \(NLP\) - GeeksforGeeks](#)
 - b. [0_Phases of NLP](#)
2. Text Preprocessing:
 - a. [Getting started with Text Preprocessing](#)
 - b. [1_Text Preprocessing for Machine Learning](#)
3. **Bag of Words and TF-IDF :**
 - a. [What is Bag of Words?](#)

- b.  Term Frequency Inverse Document Frequency (TF-IDF) Explained
 - c. [Understanding TF-IDF \(Term Frequency-Inverse Document Frequency\) - GeeksforGeeks](#)
4. **Word Embedding:**
- a.  Word Embeddings, Word2Vec And CBOW Indepth Intuition And Working- Part 1 | ...
 - b. [Comparison Between CBOW and Skip-Gram Models - GeeksforGeeks](#)
5. **Naive Bayes for Text:** Multinomial NB
- a. [Naive Bayes Classifiers - GeeksforGeeks](#)
 - b.  Tutorial 49- How To Apply Naive Bayes' Classifier On Text Data (NLP)- Machine Le...
 - c. **Assignment Notebook 1:** [Naive Bayes Text Classification](#)
6. **Language Model:** N-gram Model
- a. [N-gram in NLP - GeeksforGeeks](#)
 - b. **Assignment Notebook 2:** [N_gram_Models_NLP.ipynb](#)
7. **RNNs and Long-Range Dependencies:**
- a.  The Power of Recurrent Neural Networks (RNN)
 - b. [Introduction to Long Short Term Memory - GeeksforGeeks](#)
 - c. [Gated Recurrent Unit Networks - GeeksforGeeks](#)
 - d. [Understanding LSTM Networks -- colah's blog](#)
 - e. [\[1912.05911\] Recurrent Neural Networks \(RNNs\): A gentle Introduction and Overview](#)
 - f.  3_Recurrent Neural Networks & Sequential Learning
8. **Named Entity Recognition:**
- a. [Conditional Random Fields: Probabilistic Models for Segmenting and Labeling Sequence Data](#)
 - b. [Named Entity Recognition\(NER\) using Conditional Random Fields \(CRFs\)in NLP | by Mehul Gupta | Data Science in Your Pocket | Medium](#)
 - c.  Conditional Random Fields (CRF) - Explained
 - d. **Assignment Notebook 3:** [Assignment3_NER_CustomerSupport.ipynb](#)